

POLYALPHABETIC CIPHER TECHNIQUE (VIGENÈRE CIPHER)

AIM:

To implement the Polyalphabetic (Vigenère) Cipher technique for encrypting a plaintext message.

ALGORITHM:

1. Start.
2. Read the plaintext.
3. Read the keyword.
4. Repeat the keyword to match the plaintext length.
5. Encrypt each character using the corresponding keyword character.
6. Display the ciphertext.
7. Stop.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char text[100], key[100];
```

```
    int i, j = 0;
```

```
    printf("Enter Plain Text (Uppercase): ");
```

```
    scanf("%s", text);
```

```
    printf("Enter Key (Uppercase): ");
```

```
    scanf("%s", key);
```

```
    int keyLen = strlen(key);
```

```

printf("Cipher Text: ");

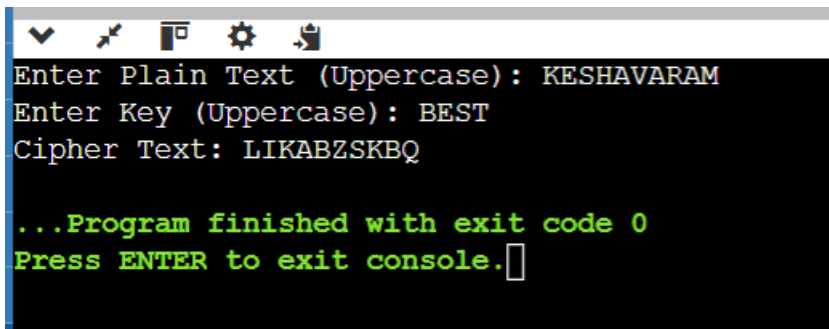
for(i = 0; text[i] != '\0'; i++)
{
    char ch = ((text[i] - 'A') + (key[j] - 'A')) % 26 + 'A';
    printf("%c", ch);

    j++;
    if(j == keyLen)
        j = 0;
}

return 0;
}

```

OUTPUT:



The screenshot shows a Windows command prompt window with a black background and white text. The window title bar is visible at the top. The text displayed is as follows:

```

Enter Plain Text (Uppercase): KESHAVARAM
Enter Key (Uppercase): BEST
Cipher Text: LIKABZSKBQ

...Program finished with exit code 0
Press ENTER to exit console.

```

RESULT:

The Polyalphabetic (Vigenère) Cipher encryption was implemented successfully.

